Exploring User Perceptions of a Campus Guide Robot: A Comparison with Human Help Desks

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Abstract

Navigating large university campuses can be challenging for newcomers and visitors, often leading to confusion or reliance on limited human help desks. This paper proposes an exploratory mixed-methods study to examine how a social robot can serve as a campus guide to support wayfinding while fostering inclusion, independence, satisfaction, and trust. Using the Misty II robot, the study would compare user experiences with robot-assisted navigation and traditional human assistance. Participants would interact with the robot in real campus environments through a combination of scripted and open-ended tasks. Observations, surveys, and short interviews would capture both behavioral and attitudinal responses. Through this proposed study, we aim to identify patterns in how people might experience and perceive the robot's guidance and to understand what design features could help make such robots feel approachable, inclusive, and trustworthy. The findings would help inform future development of social robots that complement human help systems and enhance accessibility in campus environments.

Keywords

Human-Robot Interaction, Campus Guide Robots, Inclusion, Independence, Satisfaction, Trust

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1 Introduction

Finding one's way around large university campuses can be a challenge for newcomers, visitors, and even returning students. Buildings are often spread far apart, and it is not always obvious where help desks are located or whether anyone is available to assist. This everyday challenge points to a growing need for accessible, reliable, and inclusive navigation support. In recent years, guide or navigation robots have started to emerge as one possible solution. These robots combine movement, spatial awareness, and conversation to help people reach their destinations while making the experience more personal and intuitive.

Social robots have already been used in places like museums, malls, and airports to guide people and answer questions. They not only provide directions but also create small moments of connection through gestures, voice, and presence. However, on university campuses, research on such robots remains limited. Most institutions still rely on static maps, signs, or human help desks, which do not always offer consistent accessibility for everyone. This gap raises an interesting question: could a campus guide robot help people feel more included and independent as they navigate a new environment?

This proposal explores that question by outlining a study comparing people's experiences with a campus guide robot and with traditional human help desks. The focus is on how these interactions would shape users' feelings of inclusion, independence, satisfaction, and trust. These constructs help to understand not just performance, but how supported participants feel during guidance.

The goal is to assess whether a robot might provide a similar or better sense of support compared to a human guide. Could it make navigation easier, faster, or less stressful? Or would users miss something important that humans uniquely provide?

Since there is little existing research on university navigation robots, this proposed study takes an exploratory approach. We would observe how people naturally interact with Misty II, a small, expressive robot, in real campus environments across two conditions: participants would receive wayfinding assistance from the robot in one session and from a human help desk assistant in another, counterbalanced session. Both scenarios would involve identical navigation tasks at identical campus locations, enabling direct comparison between robot and human guidance. Following both sessions, participants would complete surveys, be observed, and take part in short interviews to give detailed insight into approachability, inclusivity, and trust.

2 Related Work

2.1 Campus Guide Robots and User Perceptions

In recent years, numerous studies have investigated how social robots can assist individuals in navigating and interacting in public spaces, including campuses, museums, and shopping malls. These robots are designed to provide information, guide people, and build positive user experiences through voice, gesture, and adaptive behaviours. Across these studies, the main goal is to improve human–robot interaction (HRI) by designing robots that feel natural, understandable, and trustworthy.

A study focused on a campus guide robot designed to assist students in navigating university spaces and answering questions

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about campus facilities. The robot was built using the LoCoBot WX250S platform and used large language models (LLMs) to process user queries, break them into actions, and respond conversationally. It combined speech recognition modules for different accents and tones, object detection, and mapping algorithms to enable safe navigation in crowded areas. The system employed a semi-autonomous design, allowing a remote human operator to monitor and take control if the robot encountered difficulties or confusion. User responses showed that students found the robot helpful for wayfinding and appreciated its ability to switch between speech and gesture when explaining directions. The study emphasised that maintaining fault tolerance, safety, and natural dialogue flow were key factors in user acceptance and satisfaction [14].

NarraGuide [15], extended the idea of guide robots into museum and tour environments, focusing on narrative-based interaction. Instead of only giving directions, the robot told stories and explained exhibits in a simple, conversational way. The system allowed both robot-led and user-controlled exploration. Users could give commands, such as "go to exhibit five," or let the robot decide what to show next. The design used semantic maps, SLAM navigation, and a language model-based dialogue system that generated contextual stories linked to nearby exhibits. Some users saw it as a guide helping them learn about the museum, others as a tool or medium that extended their presence in a remote space, and a few saw it as a social companion. However, some users felt restricted when the robot took initiative without confirmation, and a few struggled with unclear references when the robot mentioned multiple objects simultaneously. The study also found that users tended to trust the robot's knowledge, even when it was inaccurate, indicating that design elements such as tone, confidence, and responsiveness significantly influence user trust. Overall, the NarraGuide project emphasised that striking a balance between autonomy, clarity, and shared control is crucial in fostering meaningful and trustworthy interactions between humans and robots.

Another study by Kanda et al. (2010) introduced a communication robot in a shopping mall designed to provide information, guide customers, and build rapport through repeated interactions. The robot, Robovie-IIF, used floor sensors, RFID tags, and cameras to detect people and identify returning visitors. It combined autonomous behaviours with remote operator support, a network robot system, to overcome real-world challenges like background noise, unpredictable questions, and social cues. The robot greeted users by name, remembered past conversations, and used gestures such as pointing and nodding to improve communication. Over a 25-day field trial, it interacted with over 2,600 visitors, many of whom described it as friendly, intelligent, and emotionally expressive. Repeat visitors developed a sense of familiarity and enjoyment, and the study even showed that the robot influenced shopping decisions. The study showed that when robots build rapport and communicate in a more personal and emotional way, users tend to feel closer to them. This sense of connection makes people more willing to interact and follow guidance, improving both satisfaction and task success [3].

2.2 Inclusion, Trust, Privacy, and Satisfaction

Designing social robots for public places means prioritising inclusion, trust, privacy, and satisfaction, and letting actions speak louder than words. speech systems that accommodate multiple accents and languages, straightforward phrasing, and gentle prosody help more people feel seen. The DRAGON guide robot works well for people with visual impairments, it utilises free-form speech, open-vocabulary landmark recognition, and dialogue to clarify goals, describe nearby objects, and maintain user orientation, features that users rated as intuitive and helpful for "gaining awareness of the environment." [8]. This idea can be generalised to public space robots, good robots "fit the place." Placemaking encourages designers to connect design with local culture and daily life, observing how people use spaces and creating environments that welcome both primary users and those simply passing through [9, 10].

Trust grows from reliable, considerate behaviour in real tasks. For example, privacy-aware navigation (PANav) utilises vision-language models to select routes that bypass busy areas when transporting sensitive items, prioritising discretion over the shortest path [11]. That kind of steady, context-aware performance signals competence and care, which people read as trustworthy [12]. Studies on robots in public spaces reveal gaps in privacy standards and call for stronger safeguards, including limits on recording, encryption of stored data, compliance with regulations like GDPR, and clear consent options and emergency stop features to prevent excessive data collection. Frameworks also stress respectful distance, minimal exposure in routing, and dataset practices that anonymise people [5, 11, 12].

Satisfaction should be measured both directly and indirectly. Direct methods include short, validated questionnaires on comfort, safety, clarity, helpfulness, and trust (plus open comments), administered after people have actually used the system. Current evaluation guidelines recommend reporting both "social" ratings (e.g., comfort, legibility) and classic navigation metrics. Indirect signals can include repeat visits, completion rates, and interaction logs; in the wild, bystanders' smooth "granting of passage" is itself a positive cue [12]. A stationary guide robot handled thousands of questions (most commonly "Where's the restroom?"), used pointing gestures and built rapport through repeated interactions.

The study also revealed novelty effects and highlighted the need for more personalised and context-aware knowledge over time [3]. Building robots that truly fit into human spaces is not just about technology; it's about empathy. Robots need to understand and adapt to the people around them, speaking in ways that feel natural, moving with awareness of personal boundaries, and respecting privacy as much as they value efficiency. When robots consistently act with care, reliability, and fairness, people start to trust them not because they're machines, but because they behave responsibly. And when users feel seen, safe, and understood, satisfaction naturally follows [5, 10].

2.3 Social Robots and Comparative User Studies

Comparative studies in public-space HRI show that guide robots and human help desks are judged on overlapping things clarity and speed, social warmth, safety/privacy, and they're evaluated both directly (short scales like Godspeed, RoSAS/HRIES, UTAUT,

SERVQUAL) and indirectly (task success, repeat use, dwell time, bystander "granting passage"). Godspeed/RO-SAS/HRIES and trust models are standard for perceptions and trust, while UTAUT or SERVQUAL cover usefulness and service quality [1, 2].

In malls, a stationary "communication robot" handled 2,642 real interactions (most often wayfinding questions, such as where to find restrooms); logs and brief questionnaires captured recognition of the robot's role, guiding behaviour, and rapport over repeat visits, providing evidence that robots can quickly and efficiently satisfy routine information needs [3]. DRAGON's free-form dialogue and open-vocabulary landmarking helped people, including visually impaired users, become more aware of their surroundings. Users found these features intuitive and helpful, enhancing their sense of independence [8].

Trust primarily comes from a robot's performance and reliability, rather than its appearance [4]. Consistent and transparent behaviour makes the biggest difference [4]. Robots should avoid unnecessary recording, encrypt any stored data, respect consent, and plan routes and behaviours that limit exposure, as reflected in SRPS guidance and privacy-aware navigation research [5]. In terms of head-to-head service quality, robots often match or surpass humans in reliability/responsiveness for simple tasks, but still fall short in empathy [1]. Therefore, tone, word choice, and considerate spacing are crucial. Finally, recent reviews argue that personalisation boosts experience but raises ethical/privacy concerns, and that trust should be measured consistently across studies, providing useful guidance for our campus Robot [6, 7]. In this study, we aim to explore how a campus guide robot can shape user experiences in real navigation tasks. Specifically, we ask the following research question:

Research Question. *How does interacting with a campus guide robot influence users' perceptions of inclusion, independence, satisfaction, and trust, compared to receiving assistance from human help desks or guides?*

3 Method

This proposed study would use a mixed-methods approach to understand how interacting with a campus guide robot might influence people's feelings of inclusion, independence, satisfaction, and trust. Mixed methods capture both the depth of user experiences through qualitative feedback and measurable patterns in perceptions through surveys, reflecting best practices in HRI research [16].

3.1 Robot Design

The study would use Misty II, a developer-friendly social robot that can create 3D maps of its surroundings, move autonomously, recognize faces, interpret voice commands, and express itself through an LCD face. Misty II is equipped with a depth sensor for mapping, a high-resolution camera for facial and object recognition, and multiple microphones to detect voice. It also includes capacitive touch panels for simple physical interactions.

The robot's tread-based system allows smooth navigation on indoor surfaces. It can autonomously return to its charging dock when needed, enabling long-duration operation. The robot would be programmed in JavaScript for navigation and task handling. Misty's expressive behaviors make it a strong candidate for supporting social, approachable navigation tasks in public spaces.



Figure 1: Misty II Robot

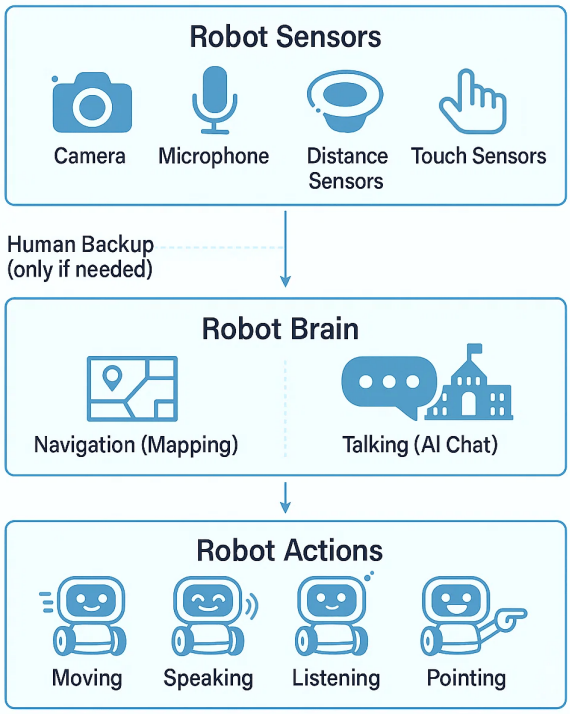


Figure 2: Campus Guide Robot System Design

3.2 Participant Sample

The participant sample would comprise college students and campus visitors needing navigation assistance. Since the study emphasizes qualitative richness, theoretical saturation would serve as the recruitment criterion. Participants would continue to be enrolled until no new themes emerged from the qualitative data [16]. We

anticipate reaching saturation at approximately 12–16 participants, providing adequate variation for basic quantitative analysis as well. Most participants would likely be students, but the expected outcomes could also extend to other visitors like prospective students, staff, or family members. The primary unit of analysis would be the dyad: the interaction between each participant and either the robot or human during the navigation task. Each person would experience both conditions in counterbalanced order.

3.3 Procedure

The study would take place on the university campus. Participants would complete two sessions in randomized order: (1) robot-assisted wayfinding and (2) human-assisted wayfinding. Each session would include identical navigation tasks, such as locating campus buildings like the library or registrar's office. Both scripted and open-ended navigation prompts would be used to observe consistency and spontaneity in participant responses.

For the robot condition, the Misty II robot would primarily operate in a semi-autonomous mode, utilizing SLAM-based pose localization and mapping to navigate smoothly and safely across campus environments. The robot would generate dynamic, context-aware verbal responses by integrating retrieval-augmented generation (RAG) with large language models (LLMs), enabling flexible and natural interactions with participants.

A Wizard-of-Oz (WoZ) setup would be employed as a fallback mechanism, used only in exceptional cases such as system failures, navigation difficulties, or to assure reliable performance during testing phases. This hybrid semi-autonomous/WoZ approach balances technological autonomy with practical safety and usability considerations in real-world settings.

In the human condition, a trained assistant would provide equivalent guidance through verbal and gestural interaction.

Each session would start with informed consent and a demographic form. Before the tasks, participants would complete a pre-interaction survey capturing baseline expectations and familiarity with robots. Observers would take detailed notes on engagement, hesitation, body language, and spoken interactions during the task. Afterward, participants would complete post-interaction surveys, followed by a 10–15 minute semi-structured interview to reflect on experiences and provide open-ended feedback.

3.4 Data Collection

Data would be collected through three complementary sources designed to triangulate quantitative and qualitative findings [16]:

Surveys: Participants would complete pre- and post-interaction surveys for both conditions. All survey items would use 7-point Likert scales.

For *Inclusion*: items adapted from the Social Connectedness Scale (e.g., "I felt welcomed by the guide," "I felt the guide was interested in helping me").

For *Independence*: items assessing capability and confidence (e.g., "I felt confident finding my way with this guide's help," "The guide's assistance made me feel capable").

For *Satisfaction*: items measuring enjoyment and utility (e.g., "I enjoyed interacting with this guide," "The guide met my expectations for assistance").

For *Trust*: items measuring reliability and understanding (e.g., "I trusted the guide's directions," "I was confident in the guide's ability").

In addition, participants would rate warmth and competence using the Godspeed questionnaire series (e.g., "friendly–unfriendly," "intelligent–unintelligent"), validated in HRI studies [16].

Observational Notes: Trained observers would record behavioral details such as gestures, tone, hesitations, engagement, and nonverbal cues that enrich interpretation of satisfaction, inclusion, and trust.

Semi-structured Interviews: Immediately following each condition, brief interviews would allow participants to describe experiences in their own words. They would discuss what made the interaction helpful or confusing and compare robot and human guidance. Recordings would later be transcribed for analysis.

3.5 Analysis

Both qualitative and quantitative analyses would be used to interpret differences between conditions.

Quantitative analysis: Survey responses would be evaluated using descriptive statistics (means, standard deviations, distributions) per construct. Comparisons between robot and human conditions would use non-parametric Wilcoxon signed-rank tests due to small sample size and non-normality assumptions [16]. Significance would be assessed at $\alpha = 0.05$, and effect sizes ($r = \frac{Z}{\sqrt{N}}$) would be included for interpretability.

Qualitative analysis: Interview transcripts and observational notes would undergo thematic analysis. Two coders would independently derive codes, with a third reviewer facilitating reconciliation, and iteratively consolidate them into higher-level themes representing inclusion, independence, satisfaction, and trust. Comparisons between conditions would help identify thematic similarities and divergences, supported by illustrative excerpts.

This mixed-methods framework would provide both broad numerical evidence and deep qualitative context, yielding comprehensive insight into user perceptions of human versus robot-assisted navigation.

4 Ethical Considerations

Because Misty II is equipped with cameras and microphones capable of recording audio and video data, it is essential to address participant privacy and consent throughout the study. Prior to participation, individuals would be informed about the potential collection of audio and video recordings during the interaction, and written informed consent for such recording would be obtained. We would obtain explicit consent from participants for audio or video recording of interviews and sessions, which would be transcribed for analysis. Participants would be informed about how recordings are managed, stored, and their rights regarding data withdrawal. Even if collected data are not stored or disseminated, informing participants respects their privacy and aligns with ethical research practices [16].

During sessions involving Misty II, the robot's blue tally light would be active when capturing personally identifiable information such as video or audio, providing an additional passive notification to participants.

The study employs a primarily semi-autonomous robot control approach complemented by a Wizard of Oz (WoZ) methodology as needed, where a human operator covertly controls the robot to simulate autonomous behavior in exceptional cases. To adhere to ethical standards in HRI WoZ research, participants would be fully debriefed after their sessions, i.e., explicitly informed about any WoZ control involved and the nature of robot operation ("lifting the curtain") [16]. This disclosure aims to prevent deception-induced distress and build trust in the research process.

Participants would also retain the right to withdraw any recorded data post-debriefing. Any use of participant data would comply with institutional review board (IRB) approvals and relevant regulations.

Overall, these ethical provisions ensure transparency, respect for autonomy, and safeguarding of participant privacy throughout the study [16].

5 Conclusion

This proposed mixed-methods study aims to explore how social robots can facilitate accessible, inclusive campus navigation. By combining observational, survey, and interview data, the approach would identify design factors promoting trust, independence, and satisfaction in human–robot interaction scenarios relevant to real-world academic environments.

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